Contents

MY FIRST LaTeX DOCUMENT

PHILIP THEURI

June 8, 2026

This is my first document created offline using LaTeX.

0.1 Objectives

1. To get a better understanding of what LaTeXis.
2. be able to form and write documents using LaTeX
3. Form and edit documents. also learn the basics of using LaTeX

1 INTRODUCTION

LaTeX is used for creating professional reports, assignments, books and research paper

The added section of parident is to forbid the computer from indenting my work and making it look shoddy. ie; to stop making new paragraphs unless i authorize it)
(if you want to underline some words then use the backslash and brackets. for long sentences then add the package ulem.add in your line uline for a single line and uuline for double underlining or use backslash textbf open curly bracket backslash underline) you can use the **backslash textbf to make your words bold**, *backslash textit to make your words Italic*

2 LaTeX LEARNING

The backslash pagebreak is to be used incase you see that your listing has gone to a new page but it wold have been fit for it to be included in one paragraph hence you move it to the next page.

2.1 superscripts

$$2x^3$$

$$2x^{34}$$

$$2x^{3x+4}$$

$$2x^{3x^4+5}$$

2.2 Subscripts

$$x_1$$

$$x_{1_2}$$

$$x_{1_{2_3}}$$

$$a_0, a_1, a_2, \ldots, a_{100}$$

$$\Pi$$
$$\pi$$
$$\alpha$$

$$A = \pi^2$$

2.3 Trig functions

$$y = \sin x (correct)$$
$$y = \cos x$$

$$y = \csc \theta (correct)$$
$$y = \sin^{-1} x$$
$$y = \arcsin x$$

2.4 Logarithmic functions

$$y = \log x$$

(remember to) put the backslash always

$$y = \log_5 x$$
$$y = \ln x$$
$$y = \csc \theta$$
$$y = logx$$

(-incorrect)

$$y = \log x$$

(-correct)

2.5 Roots

$$\sqrt{2}$$
$$\sqrt[3]{2}$$
$$\sqrt{x^2 + y^2}$$
$$\sqrt{1 + \sqrt{x}}$$

2.6 Fractions

$$\frac{5}{7}$$

About $\dfrac{2}{3}$ of the glass is full.

About $\frac{2}{3}$ of the glass is full.
About $\dfrac{2}{3}$ is full.

$$\frac{\sqrt{x+1}}{\sqrt{x+2}}$$

$$\frac{1}{(1+\frac{11}{x})}$$

You can however use dfrac to make your fraction look much better as shown below

$$\frac{1}{(1+\frac{11}{x})}$$

3 CALCULUS.

3.1 brackets

3.2 curved brackets

The distributive property states that $a(b+c) = ab + ac$, for all $a, b, c \in \mathbb{R}$.
(Using the mathmode to display equations. For example i want to display a system of two equations i can use:

$$2x + y = 5$$
$$x - y = 1$$

This merely creates two separate inline equations. You can't align the equal sign,number the equations, or treat them as one mathematical block hence you have to use the amsmath in this case. the(dollar)symbol is used to line up the equal signs.

Use the single dollar sign for short inline equations eg;The area is $A = \pi r^2$
you can also use it for different cases as align, gather, cases, matrix, etc.just by using amsmath for professionalism you can use

$$x^2 + y^2 = z^2$$

to display inline math mode rather than the previous format of using

$$x^2 + y^2 = z^2$$

-they have got the same output but tidying up the work requires the square brackets)

The equivalence class of a is $[a]$

3.3 curly brackets

The curly brackets act as a reserved word for latex hence it won't print out as int he case below:
"The section a is defined to be $1, 2, 3$." you need to use a backslash before and after the curly brackets just as we a seen earlier in a similar case.

Now using the slash back;"The section a is defined to be $\{1, 2, 3\}$."

3.4 The dollar symbol

It is also a reserved symbol used to display math mode hence it can't be printed out normally if you want to print it out.
Eg."the movie ticket costs $11.50

$$2(\frac{1}{x^2-1})$$

. . . this will print out a message not well covered brackets as we would require, you can use a single $ or use

$$2\left(\frac{1}{x^2-1}\right)$$

The backslash left before the first curly bracket and back slash right before the last curly bracket will now enable the numbers to be well indented

$$2\left\{\frac{1}{x^2-1}\right\}$$

$$2\left[\frac{1}{x^2-1}\right]$$

$$2\left[\frac{1}{x^2-1}\right]$$

$$2\left(\frac{1}{x^2-1}\right)$$

for vectors using angular brackets

$$2\left\langle\frac{1}{x^2-1}\right\rangle$$

for absolute values;

$$2\left|\frac{1}{x^2-1}\right|$$

There may be cases where you could need to open only one curly bracket especially in calculus

$$\left.\frac{dy}{dx}\right|_{x=1}$$

. . .

the period takes care of the other absolute bar. If you include a backslash left then you also need a backslash right for the same and that's where the period comes in.

you could also have different brackets being displayed in the same equation.

$$\left(\frac{1}{1+\left(\frac{1}{1+x}\right)}\right)$$

3.5 TABLES

if i want my elements in the table to be centered then i will use

$$(cccc)$$

and if i want them to appear on the right i will use

$$(lll)$$

. No of letters you enter represents the no of columns that you wanna add.

if you want to add caption in tables or images then use the backslash caption curly brackets your caption. You can also mix the letters to suit your data This is often useful because text usually looks best left-aligned, while numbers often look better centered or right-aligned. Use the backslash hline to draw borders in table

Table 1: these values represents the $f(z)$

x	1	2	3	4	5
$f(z)$	10	11	12	13	14

Table 2: These values represents the function $f(y)$

x	1	2	3	4	5
$f(y)$	$2 \cdot \dfrac{1}{x^2}$	11	12	13	14

The single dollar single dollar sign worked out because we wanted inline math mode and not display math mode. the display math mode covers a whole line .the best formulas for both inline math mode and display math mode are;
Inline math: $\frac{3}{7}$ Display math:

$$x^2 + y^2 = z$$

The curved brackets and curly brackets are used for inline and display math mode respectively. You must tell your compiler where you want your table to be lest it changes and places the table where it seems best using the backslash table H as illustrated in the code in line (155) and end table in line (166).
This function tells the computer to place the table in a specific place close to where you placed your function and where there is enough space for a table to appear.

3.6 Sentences in a table

There may be a case where you'd want to include one or more sentences in your table.It is advisable to include them starting from the left rather than the center.
There may be a case where you may have two or more sentences that should be accommodated in the same column. In this case the column part changes from using a c to p say 2cm or 2in as written.

(NB:THE COMPUTER DOESN'T UNDERSTAND FULL WORDS HENCE YOU SHOULD KEEP IN CONSIDERATION OF THESE SHORT FORMS OR SAY KEY WORDS)

$*****$ You can also do shading in tables as well as many other things . . . to be studied later $*****$

4 EQUATIONS AF ARRAYS

never use a dollar sign in align since the compiler itself is aware that you're in math mode. arrays

$$5x^2 \text{ place your text here}$$

in math mode the computer completely ignores the spaces as you would have noticed earlier every time we entered into it. if you want the space to be noted by the computer then put a space bar after the curly

x	1	2	3	4	5
$f(x)$	$2\left(\dfrac{1}{x^2}\right)$	11	12	13	14

Table 3: these values represents the $f(x)$

Table 4: The relationship between f and f'

$f(x)$	$f'(x)$
$x > 0$	The function $f(x)$ is increasing

Table 5: The relationship between f and f'

$f(x)$	$f'(x)$
$x > 0$	The function $f(x)$ is increasing. The function $f(x)$ is increasing. function $f(x)$ is increasing.

bracket or force the computer by using a backslash comma to take care o the rest

$$5x^2 - 9 = x + 3 \tag{1}$$
$$5x^2 - x - 12 = 0 \tag{2}$$

$$\begin{aligned} 5x^2 - 9 &= x + 3 \\ 5x^2 - x - 12 &= 0 \\ &= 12 + x - 5^2 5 \end{aligned}$$

The compiler itself numbers your equations hence it becomes easy to refer your equation at a later time. In order for the equations to align themselves at the equal sign, include an amber sand before
If you do not want your equations to be numbered then use an asterix in the align bracket both for the beginning and the end

$$5x^2 - 9 = x + 3 \tag{3}$$
$$5x^2 - x - 12 = 0 \tag{4}$$

$$\begin{aligned} 5x^2 - 9 &= x + 3 \\ 5x^2 - x - 12 &= 0 \\ 13 &= 12 + x - 5^2 5 \end{aligned}$$

5 CREATING LIST

5.1 Use of numbers, bullets and letters

5.1.1 numbers

we want to number or list pour items but if you just use use normal listing then the compiler will just assume them and use one sentence instead as shown below; pencil calculator ruler notebook
using line breaks would also not be advisable as it would not output the required format
it is advisable to use a begin enumerate and end enumerate as shown below

1. pencil
2. calculator
3. notebook
4. ruler

if you want a bulleted list then change from enumerate to itemize as shown below;

- pencil
- calculator
- notebook
- ruler

5.1.2 nested lists

we can have a nested list ie; an enumerated list inside another enumerated list

1. pencil
2. calculator
3. notebook
   (a) notes
   (b) homework
   (c) assignments
   (d) assessments
      i. Random assessment tests
      ii. Continuous assessment test
      iii. Quizzes
      iv. journal entries
   (e) diagrams
4. highlighters
5. ruler

One may like to use another method of listing rather than numbers or parenthesis.For that case you will require a package in order to load properly into your compiler

A. pencil

B. calculator

C. notebook

D. ruler

1. pencil
2. calculator
3. notebook
4. ruler

i. pencil
ii. calculator
iii. notebook
iv. ruler

. pencil

. calculator

. notebook

. ruler

if you dont want your listing to start from one then you will add backslash setcounter then enumi and the number that you want your listing to begin.if you type 5, it will begin listing from 6.

Example;

6. pencil
7. calculator
8. notebook
9. ruler

if your listing looks shoddy and you want it to appear in one page then use backslash page break

- pencil
- calculator
- notebook
  - notes
  - homework
  - assignments
  - assessments
    a. Random assessment tests
    b. Continuous assessment test
    c. Quizzes
    d. journal entries
  - diagrams
- highlighters
- ruler

in a case where you do not want your list to be numbered, then use square brackets but leave them empty as shown below;

pencil

calculator

notebook

ruler

you can also customize for any letter to appear manually as shown below

one. pencil

a. calculator

23. notebook

1. ruler